PTI STUDY GUIDE - MODULE 1: MEASUREMENT TECHNIQUES

Introduction

Professional tree assessment requires accurate measurement of structural defects and decay. This module covers the tools, techniques, and interpretation methods used to quantify tree conditions. Understanding these fundamentals allows you to evaluate assessment data and make informed decisions.

Module 0 taught you tree structure and biology. Module 1 teaches you **how to measure defects** in that structure.

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1. CRACK MEASUREMENT

What Are Cracks?

Cracks are separations in wood tissue that indicate structural failure progression. They differ from:

- **Natural fissures** in bark (normal, not structural)
- **Seams** at branch unions (can be normal or problematic)
- **Wounds** from damage (entry points, not cracks themselves)

Measurement Tools

Calipers (Vernier or Digital):

- Measure crack depth by inserting jaws into opening
- Accuracy: ±0.5mm
- Best for: Accessible cracks, depth <150mm
- Limitation: Can't measure deep/inaccessible cracks

Crack Gauges:

- Graduated measuring devices inserted into crack
- Can measure deeper cracks (up to 300mm+)
- Some designs include angle measurement
- Used for tracking progression over time

Tell-Tales:

- Glass or plastic strips bridging crack

- Detects movement/widening
- Installed across crack, checked monthly
- Breakage indicates active progression

How to Measure

Step 1: Locate the crack

- Note position: vertical, horizontal, spiral
- Identify deepest point (usually mid-length)
- Check for multiple cracks (measure each)

Step 2: Measure depth

- Insert caliper/gauge perpendicular to crack face
- Measure at deepest accessible point
- Record in millimeters
- Note if crack depth couldn't be fully measured

Step 3: Measure length

- Measure visible crack extent
- Note if crack continues beyond visible area
- Record orientation (vertical/horizontal/spiral)

Step 4: Context measurements

- Measure stem/limb diameter at crack location
- This allows calculating crack as % of diameter

Interpreting Crack Measurements

Express crack depth as percentage of stem diameter:

Formula: **(Crack Depth ÷ Stem Diameter) × 100**

Example:

- Stem diameter: 400mm
- Crack depth: 80mm
- Percentage: (80 ÷ 400) × 100 = **20%**

General Significance Thresholds:

- **<15%:** Minor crack, monitor
- **15-25%:** Moderate crack, requires assessment
- **25-40%:** Significant crack, detailed evaluation needed
- **>40%:** Critical crack, urgent management required

Important caveats:

- Thresholds vary by location (union vs mid-stem)
- Crack orientation matters (perpendicular to loading = more serious)
- Progression rate is critical (stable vs advancing)
- Other factors interact (decay, location, loading)

Crack Progression Monitoring

Annual measurements track change:

- **0-5mm/year:** Stable (normal stress relief)
- **5-15mm/year:** Slow progression (monitor closely)
- **15-30mm/year:** Moderate progression (management needed)
- **>30mm/year:** Rapid progression (urgent intervention)

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2. SHELL THICKNESS (t/R RATIO)

What Is Shell Thickness?

When trees develop internal decay, a **shell** of sound wood remains around the hollow. The thickness of this shell determines structural capacity.

t/R Ratio = Shell thickness (t) ÷ Radius (R)

This ratio quantifies how much structural cross-section remains relative to the tree's size.

Why t/R Ratio Matters

A tree's resistance to bending is proportional to the **cube** of its radius. When internal decay occurs:

- Outer shell material is most structurally valuable
- Inner wood contributes less to strength
- Shell thickness determines remaining capacity

Example:

- Solid tree (radius 250mm): Full structural capacity = 1.0
- Hollow tree (shell 50mm, radius 250mm): t/R = 0.20
- Remaining capacity ≈ 32% of solid tree (simplified)

Measurement Methods

Method 1: Increment Borer

Tool: Hollow drill that extracts wood core

Process:

1. Drill from bark to center (or through cavity)
2. Extract core showing wood layers
3. Measure sound shell thickness
4. Measure radius (bark to center)

Advantages:

- Direct measurement
- Shows wood condition
- Confirms decay vs void

Limitations:

- Invasive (creates wound)
- Single measurement point
- Difficult in very large trees
- Risk if decay organisms present

Method 2: Sonic Tomography

Tool: Resistograph or acoustic tomograph

Process:

1. Place sensors around tree circumference (8-12 points)
2. Generate sound waves through stem
3. Computer analyzes wave travel time

4. Creates cross-sectional image showing decay

Advantages:

- Non-invasive
- Shows full cross-section
- Maps decay distribution
- Repeatable over time

Limitations:

- Expensive equipment
- Requires training
- Some decay types difficult to detect
- Weather/moisture affects readings

Method 3: Probing/Sounding

Tool: Steel probe or mallet

Process:

1. Probe into visible cavity openings
2. Measure depth to sound wood
3. Sound (tap) stem to detect hollow areas
4. Map cavity extent

Advantages:

- Simple, no equipment
- Quick assessment
- Can map accessible cavities

Limitations:

- Only works with accessible cavities
- Imprecise
- Can't detect closed cavities
- Subjective interpretation

Calculating t/R Ratio

Step 1: Measure total radius (R)

- From bark to center
- Use DBH tape: Diameter ÷ 2 = Radius

Step 2: Measure shell thickness (t)

- From bark to cavity edge
- Average if shell varies around circumference

Step 3: Calculate ratio

- t/R = shell thickness ÷ radius

Example:

- Tree diameter: 600mm → Radius = 300mm
- Increment borer shows: 180mm bark to center, with 90mm sound shell
- Shell thickness (t) = 90mm
- Radius (R) = 300mm
- **t/R = 90 ÷ 300 = 0.30**

Interpreting t/R Values

General Guidelines (context-dependent):

- **t/R > 0.35:** Excellent structural condition
- **t/R 0.30-0.35:** Good condition, routine monitoring
- **t/R 0.25-0.30:** Moderate concern, annual assessment
- **t/R 0.20-0.25:** Significant concern, detailed evaluation
- **t/R < 0.20:** Critical condition, urgent management

Important considerations:

- Species varies (oak more tolerant than beech)
- Tree height affects acceptable t/R (tall trees need higher ratios)
- Loading context (high wind exposure needs higher ratios)
- Decay type (brittle decay more concerning than soft decay)

- Hollow geometry (asymmetric worse than uniform)

Shell Thickness and Time

t/R changes over time as:

- Decay progresses inward (reduces shell)
- Tree grows outward (may increase shell absolute thickness but decreases ratio)
- Stress causes adaptive growth (can increase shell in response to loading)

Monitoring intervals:

- **t/R > 0.30:** Every 3-5 years
- **t/R 0.25-0.30:** Every 2-3 years
- **t/R 0.20-0.25:** Annually
- **t/R < 0.20:** Every 6-12 months or more frequently

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3. DECAY EXTENT ASSESSMENT

Why Measure Decay Extent?

Decay affects structural capacity, but the **extent** and **distribution** of decay determines actual risk. A small localized decay pocket may be less concerning than widespread decay.

Visual Assessment

External indicators:

- Fungal fruiting bodies (bracket fungi)
- Cracks or seams opening
- Stem deformity or swelling
- Bark abnormalities
- Root collar condition

Limitations:

- Decay often internal before external signs
- Fruiting bodies appear seasonally
- Can't quantify extent visually

Quantifying Decay Extent

Method 1: Percentage of Cross-Section

From tomography or bore samples:

- Measure decayed area vs total cross-section
- Express as percentage

Example:

- Total stem area: 280,000 mm^2
- Decayed area: 98,000 mm^2
- Decay extent: 35%

Interpretation:

- **<20%:** Minor decay
- **20-40%:** Moderate decay
- **40-60%:** Significant decay
- **>60%:** Extensive decay

Method 2: Radial Penetration

From increment borer:

- Measure depth of decay from cavity edge
- Express as % of radius

Example:

- Radius: 300mm
- Decay extends 120mm from cavity
- Radial penetration: 40%

Method 3: Vertical Extent

For stem/root decay:

- Measure height of decayed section
- Particularly important for butt rot

Example:

- Tree height: 18m
- Decay extends 2.5m up from base
- Vertical extent: 14% of height

Decay Distribution Patterns

Symmetrical (uniform around stem):

- More predictable failure mode
- t/R ratio most reliable
- Generally more stable than asymmetric

Asymmetrical:

- One side more affected
- Creates eccentric loading
- Failure direction more predictable
- May require lower t/R threshold

Pocket decay:

- Localized area of decay
- Can create stress concentration
- Location matters (windward vs leeward)

Root decay:

- Often asymmetric
- Affects anchorage more than stem strength
- Windthrow risk primary concern

Combining Measurements

Professional assessment integrates:

1. Shell thickness (t/R ratio)
2. Decay extent (% of cross-section)
3. Decay distribution (uniform vs asymmetric)
4. Vertical extent (how much of tree affected)

5. Loading context (wind exposure, crown size)
6. Target context (consequence of failure)

Example integrated assessment:

"Oak, 18m height, t/R 0.27, decay extent 35% concentrated on north side, root collar involvement 40% circumference, moderate wind exposure, overhangs car park."

This provides complete picture for risk assessment.

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4. CROWN CONDITION ASSESSMENT

Quantifying Dieback

Dieback = dead or dying branches in crown

Measurement method:

1. Estimate total crown volume
2. Estimate dead/dying crown volume
3. Express as percentage

Practical approach:

- View from multiple angles
- Identify dead branches (no leaves, peeling bark, fine twig loss)
- Estimate visually or use categories

Categories:

- **0-10%:** Minimal dieback (normal aging)
- **10-25%:** Moderate dieback (investigate cause)
- **25-50%:** Significant dieback (major stress)
- **50-75%:** Severe dieback (poor prognosis)
- **>75%:** Terminal decline

Crown Density

Measures foliage abundance:

- Compare to typical healthy specimen
- Express as percentage

Scale:

- **90-100%:** Excellent density
- **70-90%:** Good density
- **50-70%:** Moderate density (stress indicator)
- **30-50%:** Poor density (significant stress)
- **<30%:** Very poor density (critical condition)

Leaf Size Assessment

Chronic stress reduces leaf size:

- Compare to normal leaf size for species
- Measure sample leaves
- 50-70% of normal size indicates moderate stress
- <50% indicates severe stress

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5. LEAN MEASUREMENT

What Is Lean?

Lean = deviation from vertical (plumb)

Types:

- **Chronic lean:** Long-term, adapted (reaction wood present)
- **Recent lean:** Developed within 1-3 years (concern)
- **Progressive lean:** Actively increasing (high concern)

Measurement Tools

Clinometer (inclinometer):

- Measures angle from vertical
- Digital or analog
- Accuracy: ±0.5°

Plumb line:

- Simple weighted string
- Measure offset distance
- Calculate angle

Smartphone apps:

- Use built-in accelerometer
- Reasonable accuracy (±1°)
- Convenient but check calibration

How to Measure

Method 1: Direct angle measurement

1. Stand perpendicular to lean direction
2. Sight along stem with clinometer
3. Read angle from vertical
4. Record lean direction (compass bearing)

Method 2: Offset measurement

1. Measure height to crown center
2. Measure horizontal offset at ground
3. Calculate angle: tan(angle) = offset ÷ height

Example:

- Crown center: 12m high
- Offset: 2.5m
- tan(angle) = 2.5 ÷ 12 = 0.208
- Angle = arctan(0.208) = **11.7°**

Interpreting Lean

General thresholds:

- **0-5°:** Minimal lean (often within normal variation)
- **5-10°:** Moderate lean (assess if recent or chronic)
- **10-15°:** Significant lean (detailed assessment needed)

- **15-25°:** Severe lean (high concern)
- **>25°:** Critical lean (urgent intervention)

Critical Distinction: Recent vs Chronic

Chronic lean:

- Tree adapted with reaction wood
- May be stable for decades

Recent lean:

- Indicates failure in progress
- Much higher risk

Indicators of recent lean:

- Soil mounding (compression side)
- Root plate lifting (tension side)
- Cracks in soil
- Bark compression folds (new)
- No reaction wood (on increment core)

Monitoring Lean Progression

Track changes over time:

1. Measure initial lean (date recorded)
2. Remeasure at intervals (3-12 months)
3. Calculate rate of change

Progression significance:

- **0-1°/year:** Stable (monitor)
- **1-3°/year:** Slow progression (assess causes)
- **3-5°/year:** Moderate progression (management needed)
- **>5°/year:** Rapid progression (urgent intervention)

SUMMARY: KEY MEASUREMENT PRINCIPLES

1. Accuracy Matters

- Use appropriate tools
- Measure at consistent locations
- Record precise values (not estimates)
- Document measurement date

2. Context Is Essential

- Measurements alone don't determine risk
- Consider species, age, location, loading
- Integrate multiple measurements
- Account for environmental factors

3. Trends Over Time

- Single measurements are snapshots
- Progression rates reveal urgency
- Establish monitoring intervals
- Document changes systematically

4. Professional Standards

- Follow established protocols (BS5837, VALID, etc.)
- Consistent methodology allows comparison
- Clear documentation supports decisions
- Measurements form evidence base for management

PRACTICE EXERCISES

Exercise 1: Crack Calculation

Given:

- Tree diameter: 520mm
- Crack depth: 110mm

Question: Calculate crack as % of diameter

Answer:

- (110 ÷ 520) × 100 = **21.2%**
- **Interpretation:** Moderate crack (15-25% range), requires detailed assessment

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Exercise 2: t/R Ratio

Given:

- Tree radius: 280mm
- Shell thickness: 75mm

Question: Calculate t/R ratio

Answer:

- t/R = 75 ÷ 280 = **0.268**
- **Interpretation:** Moderate concern (0.25-0.30 range), annual assessment recommended

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Exercise 3: Lean Angle

Given:

- Tree height: 15m
- Horizontal offset: 3m

Question: Calculate lean angle

Answer:

- tan(angle) = 3 ÷ 15 = 0.2
- angle = arctan(0.2) = **11.3°**
- **Interpretation:** Significant lean (10-15° range), detailed assessment needed to determine if chronic or recent

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Exercise 4: Decay Extent

Given:

- Total stem cross-section: 450,000 mm^2

- Decayed area: 162,000 mm²

Question: Calculate decay extent as percentage

Answer:

- (162,000 ÷ 450,000) × 100 = **36%**
- **Interpretation:** Moderate decay (20-40% range), requires calculation of remaining shell thickness (t/R) to assess structural capacity

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NEXT MODULE

Module 2: Structural Assessment Tools
Covers advanced assessment equipment including sonic tomography, resistograph, pulling tests, and Visual Tree Assessment (VTA) in detail.

This study guide provides foundational knowledge for PTI training. Apply these measurement concepts when evaluating tree scenarios in practice questions.